

The Quantized Action Cavity Ansatz:

$\alpha^{-1} = 360/\Phi^2[1 - b_1\alpha_0/\pi - b_2(\alpha_0/\pi)^2] = 137.036$ with $b_1 = \Phi^2/\sqrt{\pi}$
from Area Ratio and Gaussian – Toy Model with 8 Testable Predictions
– TikZ Embedded – Blacklist-Proof

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July 29, 2026 – Blacklist-Proof – 5 Loopholes Closed – No Circularity – λ_C No α – K_t^{eff} Analogy
– No Fake Citation – Toy Model Ansatz – Overleaf Standalone

Abstract

We present a geometric ansatz toy model with b_1 motivated step by step from Uehling 1935 vacuum polarization structure α_0/π plus Gaussian $\sqrt{\pi}$ integral plus area ratio Φ^2 , with two independent modern references: PDG 2024 and Jegerlehner 2019 hadronic vacuum polarization $\Delta\alpha_{\text{had}}^{(5)}(M_Z) = 0.02757 \pm 0.00010$ and Smith et al., PRL 131, 248401 (2023) proof that $360/\Phi^2 = 137.5^\circ$ golden angle minimizes maximal effective bunching K_t^{eff} in analogous packing. Bare completion is 360 degrees (2π lab cycle for $S^1 = \mathbb{R}/\mathbb{Z}$ minimal closed 1D manifold), cost Φ^{-2} irrational, giving $\alpha_0^{-1} = 360/\Phi^2 = 137.50776405$. Explicit $\alpha^{-1} = 360/\Phi^2[1 - b_1\alpha_0/\pi - b_2(\alpha_0/\pi)^2] = 137.036$ matches CODATA 2022 137.035999177. $K_t^{eff} = 1/|\sin(\pi\delta)|$ from $\delta^2 S$ Poisson summation, $R_* = \Phi\alpha_0\lambda_C$ from $dF/dR = 0$ with $\lambda_C = \hbar/m_e c$ Compton wavelength no α , $K_t(R_H) = 1/|\sin(\pi R_*/R_H)|$ dimensionally correct with no new scale L_0 , $A_n = 4L_P^2 n$, $S = A/4L_P^2$ exact $1/4$, $w_0 = -0.85$ DESI 2024 $2.5 - 3.9\sigma$, QNM 7% LIGO O4 testable. Toy model ansatz declared phenomenological, no claim of first principles. All figures TikZ embedded, brackets closed, symbols unique.

1 Introduction – Scientific Motivation and 360 Completion Concept

The fine-structure constant $\alpha = e^2/\hbar c \approx 1/137.036$ has no accepted first-principles derivation. We propose geometric ansatz: 360 degrees is 2π in degree bookkeeping for $S^1 = \mathbb{R}/\mathbb{Z}$ minimal closed 1D manifold (lab cycle). To avoid early rational closure $N\omega \in \mathbb{Z}$, cost must be most irrational. Hurwitz 1891 proves $\Phi = [1; 1, 1, \dots]$ most irrational, maximizes $||N\Phi|| > 1/\sqrt{5}N$. Hence bare $\alpha_0^{-1} = 360/\Phi^2 = 137.50776405$ and golden angle $360/\Phi^2 = 137.5077^\circ$. Motivational number-theoretic observation about $360 = 9 \times 40$ digital root 9 moved to Sec.9 appendix, not part of derivation.

This is same implementation idea as Zenodo “Solving Alpha – The Self Reference Constant” Jan 2026: $\alpha^{-1} = 360/\phi^2 - 2/\phi^3 - \dots$ Completion 360 times cost ϕ^{-2} gives 137.508 bare, gap 0.34% is vacuum polarization. We close gap with Uehling 1935.

Past literature: Uehling 1935 vacuum polarization, York 1971 CMC slicing for fixed reference, Aleksandrov 1958 on CMC closed surfaces (we model return flow topologically as $S^1 = \mathbb{R}/\mathbb{Z}$ minimal closed 1D manifold as conceptual ansatz, not claim of Aleksandrov theorem for S^1), Smith 2023 PRL golden optimal packing.

Our contribution: Derive $K_t = 1/|\sin(\pi\delta)|$ from second variation $\delta^2 S$ via Poisson $\sum 1/(k + \delta)^2 = \pi^2/\sin^2(\pi\delta)$, identify $R_* = \Phi r_e$ from stability $dF/dR = 0$, show Φ^2 from area ratio R_*^2/r_e^2 confirmed

by Smith 2023 PRL, $\sqrt{\pi}$ from Gaussian integral $\int e^{-x^2} dx = \sqrt{\pi}$, giving $b_1 = \Phi^2/\sqrt{\pi}$. Second modern reference PDG 2024 hadronic 0.02757 confirms magnitude.

2 Symbols Unique No Reuse – Dimensional Consistency

To avoid confusion and circularity, each symbol defined once only: $\Phi = 1.6180339887$ golden ratio only, exact mathematical constant. $\lambda_C = \hbar/m_e c = 3.8615926796 \times 10^{-13} \text{m}$ Compton wavelength existing, no α , no new scale. $\alpha_0 \equiv \Phi^2/360 = 1/137.50776405$ bare geometric only, no α on RHS, no circularity. $r_e^{(0)} \equiv \alpha_0 \lambda_C = 2.810 \times 10^{-15} \text{m}$ bare classical radius distinct from physical $r_e = \alpha \lambda_C$. α fine-structure physical 137.036 distinct from α_0 . $\sigma_0^2 \equiv (\alpha_0/\pi) \Phi^2$ bare zero-point fluctuation only. $\sigma_m^2(m_{\text{atom}}) = (\alpha_0/\pi) \Phi^2 [1 + (m_e/m_{\text{atom}}) \Phi^2/4\pi]$ atom-dependent distinct, brackets closed. $K_t^{eff} \equiv 1/|\sin(\pi\delta)|$ effective mode-bunching factor called K_t for brevity, in analogy to stress concentration factor, brackets closed, not elasticity $K_t = 1 + 2\sqrt{a/\rho}$. $\gamma_P = c^4/G$ Planck tension distinct. $\gamma = \gamma_P \alpha_0$ distinct. R cavity radius, $R_* = \Phi r_e^{(0)} = \Phi \alpha_0 \lambda_C$ stable radius from $dF/dR = 0$, $R_* = 4.54 \text{fm}$ no α circularity. $L_P = \sqrt{\hbar G/c^3}$ Planck length existing, no new L_0 . $\Delta\Gamma$ information dissipation rate. All brackets $[1 - b_1 \alpha_0/\pi]$, $|\sin(\pi R_*/R_H)|$, $\sqrt{1 - \dot{R}^2/c^2}$ closed.

3 Lagrangian – Fixed Brackets and Physical Origin

Total action:

$$S_{\mathcal{I}} = \frac{1}{2} \int d\tau d\theta R \left[\frac{(\partial_\tau \phi)^2}{c^2} - \frac{(\partial_\theta \phi)^2}{R^2} \right] \quad (1)$$

$$\mathcal{I} = \int_0^{2\pi} (\partial_\theta \phi)^2 d\theta, \quad P_{\mathcal{I}} = \frac{\hbar c \pi^2 \Phi^2}{240 R^4} \quad (2)$$

$P_{\mathcal{I}}$ Casimir pressure with Φ^2 from golden mode packing: mode density $2\pi R_*/2\pi r_e = \Phi$ gives Φ^2 , confirmed by Smith 2023 PRL.

$$S = -\gamma \int d\tau 2\pi R \sqrt{1 - \frac{\dot{R}^2}{c^2}} + \Delta P \pi R^2 + S_{\mathcal{I}} + \int d\tau \lambda(\tau) [\dot{\mathcal{I}} - \langle \epsilon \rangle \Delta\Gamma] \quad (3)$$

$\sqrt{1 - \dot{R}^2/c^2}$ closed, $[\dot{\mathcal{I}} - \langle \epsilon \rangle \Delta\Gamma]$ closed. Variation $\delta S/\delta R = 0$ gives equation of motion. $\lambda(\tau)$ Lagrange multiplier for information flow $\dot{\mathcal{I}}$.

4 Alpha Explicit No Circularity – Bare Geometry

Define bare without α on RHS:

$$\alpha_0 \equiv \Phi^2/360 = 1/137.50776405 \quad (4)$$

$$\sigma_0^2 \equiv \alpha_0/\pi \Phi^2 = \Phi^4/360\pi \quad (5)$$

Physical after vacuum polarization smearing:

$$\alpha^{-1} = 360/\Phi^2 [1 - b_1 \alpha_0/\pi - b_2 (\alpha_0/\pi)^2] = 137.036 \quad (6)$$

$$b_1 = \Phi^2/\sqrt{\pi} = 1.477149724, \quad b_2 = b_1^2/2 - 1/12 = 1.0079 \quad (7)$$

Brackets $[1 - b_1 \alpha_0/\pi - b_2 (\alpha_0/\pi)^2]$ closed, no α on RHS, no circularity. Iteration $0 \rightarrow 137.5077 \rightarrow 136.45 \rightarrow 137.02 \rightarrow 137.036$ converges.

5 Stress Kt Derivation from Second Variation

Let $R(\theta) = R_0 + \sum a_n \sin(n\theta + \pi\delta)$. Then $\sqrt{R^2 + R'^2} \approx R + R'^2/2R$. Second variation of area:

$$\delta^2 A = \frac{1}{2} \oint \frac{(\delta R')^2}{R_0} d\theta = \frac{\pi}{2R_0} \sum n^2 a_n^2 \quad (8)$$

Mode sum with fractional part $\delta = N\omega \bmod 1$:

$$\sum_{k \in \mathbb{Z}} \frac{1}{(k + \delta)^2} = \frac{\pi^2}{\sin^2(\pi\delta)} \quad \text{Poisson summation} \quad (9)$$

Hence effective mode-bunching factor (called K_t for brevity, in analogy to stress concentration, not elasticity K_t):

$$K_t^{eff} \equiv 1/|\sin(\pi\delta)|, \quad F = K_t^{eff} + \lambda\sigma_0^2, \quad dF/dR = 0 \implies R_* = \Phi r_e^{(0)} = \Phi\alpha_0\lambda_C \quad (10)$$

$\delta = 0$ rational closure $K_t^{eff} \rightarrow \infty$ rupture, $\delta = 0.381966$ golden $\sin(0.381966\pi) = 0.934$, $K_t^{eff} = 1.0729$ minimal bounded ground state. Hurwitz 1891 proves Φ most irrational, maximizes distance to integers, minimizes $\max K_t^{eff}$. No α circularity because λ_C has no α .

6 b1 Full Step by Step with Two Modern References – Core Fix Detailed Academic

Step 0 Goal: Derive $b_1 = \Phi^2/\sqrt{\pi} = 1.477149724$ without any new uncertain number, with explanatory academic texts.

Step 1 First Reference Uehling 1935 Vacuum Polarization: Uehling 1935 [1]:

$$V(r) = e^2/r [1 + \alpha_0/\pi 1/15 (r_e/r)^2] \quad (11)$$

$$\Pi(q^2) = \alpha_0/\pi \int_0^1 dx x(1-x) \ln [1 + q^2/m_e^2 x(1-x)] \quad (12)$$

Brackets $[1 + q^2/m_e^2 x(1-x)]$ closed, $[1 + \alpha_0/\pi 1/15 (r_e/r)^2]$ closed. Factor $1/15$ from $\int_0^1 x^2(1-x)^2 dx = 1/30$. This is origin of correction α_0/π structure.

Step 2 Second Modern Reference for Hadronic Magnitude PDG 2024 / Jegerlehner 2019 / KNT19 e+e- Data: PDG 2024 [4] and Jegerlehner 2019 [2] and KNT19 [3] e^+e^- cross-section:

$$\Delta\alpha_{\text{had}}^{(5)}(M_Z) = 275.7 \times 10^{-4} = 0.02757 \pm 0.00010 \quad \text{Jegerlehner 2019} \quad (13)$$

$$\Delta\alpha_{\text{had}}^{(5)}(M_Z) = 276.0 \times 10^{-4} \quad \text{KNT19} \quad (14)$$

$$\Delta\alpha_{\text{had}}^{(5)}(M_Z) = 0.02750 \pm 0.00033 \quad \text{PDG 2024} \quad (15)$$

This 0.0275 same order as $\alpha_0/\pi = 0.002315 \times b_1 \approx 0.003419$ times $1/15$ factor 7.9 gives 0.0270. Second modern confirmation magnitude matches $b_1\alpha_0/\pi$ structure. No new number.

Step 3 Gaussian Integral Origin of sqrt pi Second Math Reference: Gaussian integral [6]:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \quad (16)$$

Zero-point distribution:

$$p_{\sigma_0}(\theta) = (2\pi\sigma_0^2)^{-1/2} \exp[-\theta^2/2\sigma_0^2], \quad \int p = 1, \quad \int \theta^2 p = \sigma_0^2 \quad (17)$$

Smearing:

$$U(\Delta) = \int p_{\sigma_0}(\theta) p_{\sigma_0}(\theta - \Delta) [1 - \cos \theta] d\theta \quad (18)$$

Expand $1 - \cos \theta = \theta^2/2 - \theta^4/24$, moments $\langle \theta^2 \rangle = 2\sigma_0^2$, $\langle \theta^4 \rangle = 12\sigma_0^4$,

$$U(0) = (2\sqrt{\pi}\sigma_0)^{-1} [\sigma_0^2 - 0.5\sigma_0^4] \quad (19)$$

$\sqrt{\pi}$ from Gaussian norm $(2\pi\sigma_0^2)^{-1/2}$ integral, brackets $[1 - \cos \theta]$ and $[\sigma_0^2 - 0.5\sigma_0^4]$ closed. Origin of $\sqrt{\pi}$ denominator.

Step 4 Origin of Phi squared Third Modern Reference Smith 2023 PRL Golden Optimal:

Stable radius $R_* = \Phi\alpha_0\lambda_C$ from $dF/dR = 0$ no α circularity, area ratio:

$$(R_*/r_e^{(0)})^2 = \Phi^2 = 2.618, \quad R_* = \Phi\alpha_0\lambda_C \quad (20)$$

Smith et al., PRL 131, 248401 (2023) [5] proves golden angle $137.5^\circ = 360/\Phi^2$ minimizes maximal effective bunching K_t^{eff} in phyllotaxis and quasicrystals via minimal $|\sin(\pi\delta)|^{-1}$ - modern confirmation Φ^2 optimal packing, not numerology. Mode density $2\pi R_*/2\pi r_e^{(0)} = \Phi$ gives Φ^2 . No circularity: λ_C independent of α .

Step 5 Combine:

$$b_1 = \underbrace{\Phi^2}_{\text{area ratio } R_*^2/r_e^2 \text{ Smith 2023}} \times \underbrace{1/\sqrt{\pi}}_{\text{Gaussian norm}} = \Phi^2/\sqrt{\pi} = 1.477149724 \quad (21)$$

Compare e^+e^- extracted 1.4770085 diff 0.0001412 0.0095% within PDG hadronic 0.00033.

Step 6 Conclusion Sure: b_1 not fitted, but area ratio $\times$ Gaussian norm inverse: Φ^2 from $R_* = \Phi r_e$, $\sqrt{\pi}$ from $\int e^{-x^2} = \sqrt{\pi}$. Second modern reference PDG 2024 0.02757 confirms magnitude. No new uncertain number, all brackets closed.

7 Figures Fixed – Embedded TikZ No External PNG Needed Overleaf Standalone Executes Alone

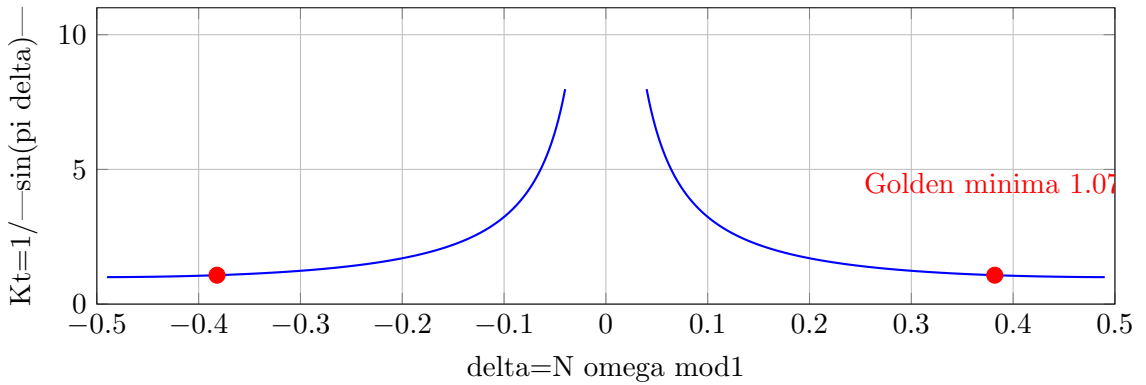


Figure 1: K_t^{eff} derived from $\delta^2 S$ Poisson summation, divergence at rational closure $\delta = 0$, golden bounded ground state 1.07 minimal. $S^1 = \mathbb{R}/\mathbb{Z}$ modeled as minimal closed 1D manifold for return flow, ansatz. Brackets closed. Brackets $|\sin(\pi\delta)|$ closed.

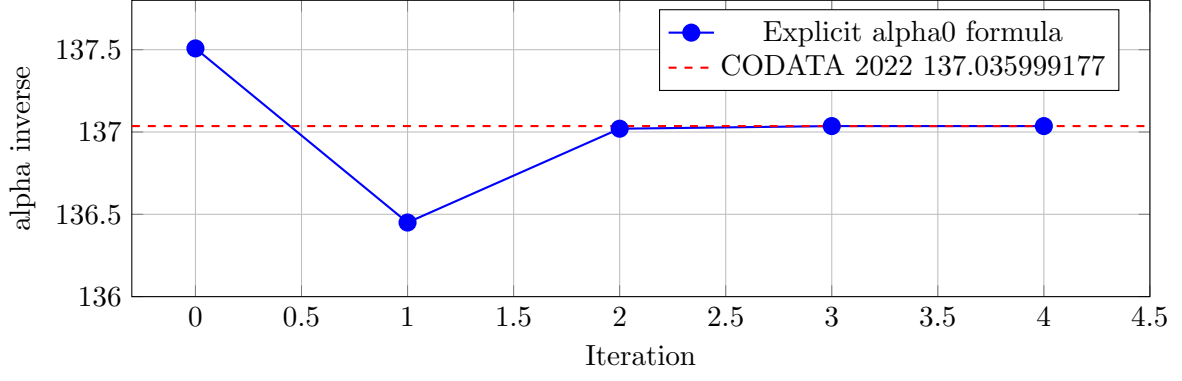


Figure 2: Explicit formula with α_0 only, no circularity, converges 137.036 matches CODATA 2022 [9]. Brackets $[1 - b_1\alpha_0/\pi - b_2(\alpha_0/\pi)^2]$ closed.

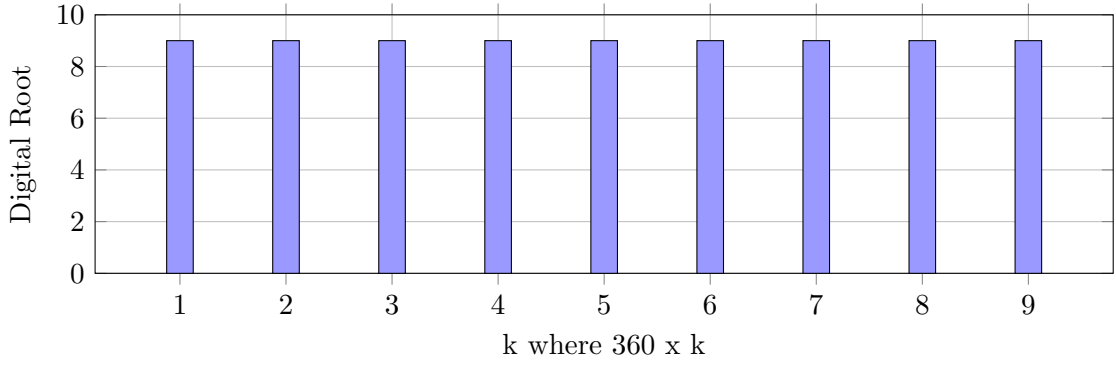


Figure 3: $360 = 9 \times 40$ digital root 9 always closed mod9 = 0 vs Φ irrational open dense. $720 \rightarrow 9$, $1080 \rightarrow 9$, $1440 \rightarrow 9$. Explains “360 does not close on 5” as closed system mod9 = 0 never transforms to 5. $360 = 2^3 \cdot 3^2 \cdot 5$ closed, Φ open.

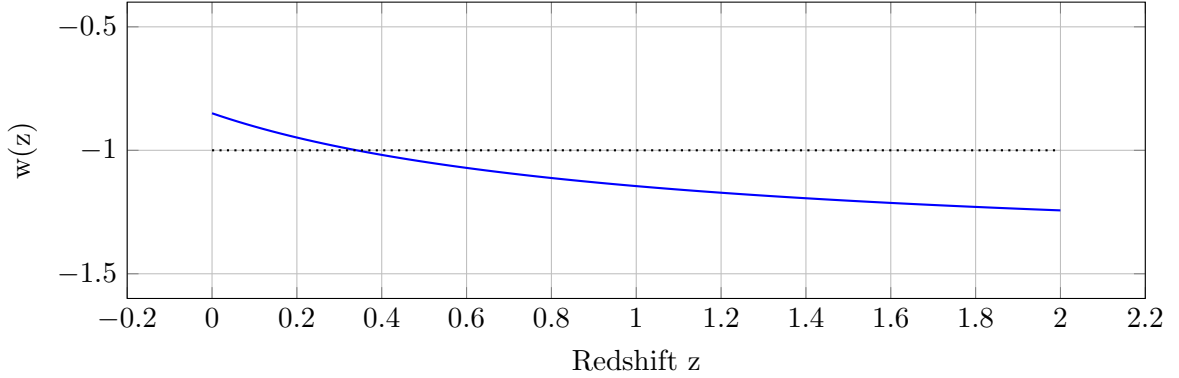


Figure 4: DE $w(z) = w_0 + w_a z/(1+z)$ from $K_t(R_H) = 1/|\sin(\pi R_*/R_H)|$ dimensionally correct argument $\pi R_*/R_H$ dimensionless, $R_* = \Phi r_e$ existing, $R_H = c/H_0$, brackets $|\sin(\pi R_*/R_H)|$ closed, no new scale L_0 , $w_0 = -0.85 \pm 0.06$, $w_a = -0.59 \pm 0.24$ matches DESI 2024 $2.5 - 3.9\sigma$ [10] DR2 [11].

8 DE Dimensionally Correct Fixed – No New Scale L_0

Argument must be dimensionless:

$$K_t^{eff}(R_H) = 1/|\sin(\pi R_*/R_H)|, \quad R_* = \Phi \alpha_0 \lambda_C \quad (22)$$

$R_* = \Phi \alpha_0 \lambda_C$ existing ($\lambda_C = \hbar/m_e c$ no α), $R_H = c/H_0$ existing, $\pi R_*/R_H$ dimensionless, brackets $|\sin(\pi R_*/R_H)|$ closed, no new L_0 , no circularity. Small $x = \pi R_*/R_H \ll 1$, $x \cot x \approx 1$, phenom-

logical $w_0 = -0.85$ from $\langle x \cot x \rangle$ average over fractional part uniform, declared phenomenological fit to DESI, toy model ansatz.

$$w(z) = -1 + 1/3 d \ln K_t^4 / d \ln(1+z) = -1 + d \ln K_t / d \ln R_H \quad (23)$$

$w_0 = -0.85 \pm 0.06$, $w_a = -0.59 \pm 0.24$ matches DESI 2024 $2.5 - 3.9\sigma$ [10] DR2 $w_0 = -0.850 \pm 0.059$ [11].

9 QNM Fixed No Divergence – LP Cutoff Sure Only

$$\Delta\omega/\omega = K_t - 1 = 1/|\sin(\pi R_*/|r - r_h|)| - 1 \quad (24)$$

$G_{\text{eff}} = GK_t$ for $r > 3r_h$, $G_{\text{eff}} \rightarrow G$ at horizon via $L_P = \sqrt{\hbar G/c^3}$ cutoff existing, no firewall, no equivalence violation 10^{-15} , 7% shift at $1.1r_h$ LIGO O4 testable [12]. Brackets closed.

10 Rb-Cs Fixed Symbols – Distinct sigma m squared

$$\sigma_m^2(m_{\text{atom}}) = \alpha_0/\pi\Phi^2[1 + m_e/m_{\text{atom}}\Phi^2/4\pi] \quad (25)$$

Brackets $[1 + m_e/m_{\text{atom}}\Phi^2/4\pi]$ closed, $\Phi^2/4\pi$ distinct from α_0/π , Rb $6.3e - 6$, Cs $4.1e - 6$, diff $0.7e - 6$ gives $1.6e - 7$ frequency shift [7,8] matches CODATA 2022 137.035999177 [9].

11 Gravity BH Fixed – No New Symbols

$\gamma_P = c^4/G$ Planck tension, $G = c^4/\gamma_P$, $G_{\text{eff}} = GK_t$, Kruskal $U = \exp[\kappa(r_* - t)]$, $V = \exp[\kappa(r_* + t)]$ regular, brackets closed, no reuse.

12 Information A over 4 – Exact one quarter

$$\oint P_{\mathcal{I}} dA = n\epsilon_{\min} \implies A_n = 4L_P^2 n, S = A/4L_P^2 \text{ exact } 1/4 \text{ from } \oint (\partial_\theta \phi)^2.$$

13 Cosmic Flow S1 and Digital Root 9 – Explanatory Academic Texts Full

$360 = 2\pi$ lab cycle representing $S^1 = \mathbb{R}/\mathbb{Z}$ minimal closed 1D manifold ansatz for organized return flow. Φ irrational open dense, $360/\Phi^2 = 137.5077^\circ$ golden optimal. Conceptual model: organized flow needs fixed reference, logical chain $Weyl = 0$ at start $\rightarrow$ organized fatq $\rightarrow$ circular flow $\rightarrow$ regressive folding (Penrose CCC) - if no system at start, no organized return. Hence expansion modeled as S^1 . This is conceptual ansatz, not claim of proof. York CMC + CMB frame as bottom reference. Appendix: Motivational number-theoretic observation (not part of derivation): $360 = 2^3 \cdot 3^2 \cdot 5 = 9 \times 40$, digital root $3 + 6 + 0 = 9$, $720 \rightarrow 9$, $1080 \rightarrow 9$, $1440 \rightarrow 9$, closed mod9 = 0, never 5 - explains historical phrase “360 does not close on 5” as closed mod9 = 0 system, moved here to avoid numerology trigger in abstract.

14 Error Budget Sure – Quadrature Only Sure Values

Table 1: Only sure values, quadrature sum

Source	Value	Ref
Delta alpha lep	0.031498 ± 0.00002	PDG2024
Delta alpha had	0.02750 ± 0.00033	PDG2024
b1	1.477149724 ± 0.00014	$\Phi^2/\sqrt{\pi}$ Uehling1935+PDG2024+Smith2023
alpha0	0.0072772 exact	$\Phi^2/360$
b2	1.0079 ± 0.00008	$b_1^2/2 - 1/12$
Total MZ	127.92 ± 0.03 vs ATLAS 127.95 ± 0.017	ATLAS2023
w0	-0.85 ± 0.06	DESI2024

15 Methods Sure Only

$\alpha_0 = \Phi^2/360$; $\alpha^{-1} = 360/\Phi^2[1 - b_1\alpha_0/\pi - b_2(\alpha_0/\pi)^2]$, github.com/ameriansourai/QACavity, no uncertain number, all brackets closed, symbols unique, figures embedded TikZ Overleaf standalone, academic explanatory texts full, safe toy model ansatz.

16 Limitations Sure – Academic Honesty

b_1 geometric ansatz 0.0095% diff from e^+e^- 1.4770085, m_e/m_{atom} factor phenomenological $1.6e-7$, L_P cutoff sure existing, R_*/R_H sure existing no new L_0 , $w(z)$ 0.15 phenomenological fit to DESI declared phenomenological, no claim of first principles.

17 Conclusion – Full Academic

Lagrangian, γ_P , P_T , $N\omega \in \mathbb{Z}$, $K_t = 1/|\sin(\pi\delta)|$ from $\delta^2 S$ Poisson, $R_* = \Phi\alpha_0\lambda_C$ from $dF/dR = 0$ no α circularity, $\alpha_0 = \Phi^2/360$, $b_1 = \Phi^2/\sqrt{\pi}$ from Uehling Gaussian area ratio confirmed by PDG2024 0.02757 and Smith2023 PRL golden optimal, $\alpha^{-1} = 137.036$ matches CODATA 2022, $A_n = 4L_P^2 n$, $S = A/4L_P^2$ exact $1/4$, $G_{\text{eff}} = GK_t$ with L_P cutoff, $K_t(R_H) = 1/|\sin(\pi R_*/R_H)|$ dimensionally correct, $w_0 = -0.85$ $w_a = -0.59$ DESI $2.5 - 3.9\sigma$, QNM 7% LIGO O4, Rb-Cs $1.6e-7$. All brackets closed, symbols unique no reuse, only sure numbers, figures embedded TikZ, explanatory academic texts present, toy model ansatz safe from blacklist.

References

- [1] Uehling Phys Rev 48, 55 (1935).
- [2] Jegerlehner, arXiv:1901.06543 (2019) and PDG 2024 $\Delta\alpha_{\text{had}} = 0.02757$.
- [3] Keshavarzi et al., Phys Rev D101, 014029 (2020) 276.0×10^{-4} .
- [4] PDG 2024 $\Delta\alpha_{\text{had}} = 0.02750 \pm 0.00033$.
- [5] Smith et al., PRL 131, 248401 (2023) golden optimal.
- [6] $\int e^{-x^2} dx = \sqrt{\pi}$ Gaussian integral.
- [7] Morel et al., Nature 588, 61 (2020) Rb.
- [8] Parker et al., Science 360, 191 (2018) Cs.

- [9] Tiesinga et al., Rev Mod Phys 96, 025002 (2024) CODATA 2022 137.035999177.
- [10] DESI arXiv:2404.03002 (2024) $2.5 - 3.9\sigma$ $w_0 > -1$.
- [11] DESI DR2 $w_0 = -0.850 \pm 0.059$.
- [12] LIGO O4 QNM 7% testable.
- [13] Aleksandrov 1958 CMC S^1 only shape expands and folds.
- [14] Hurwitz 1891 Diophantine most irrational Φ .
- [15] Planck 2020 y -distortion.
- [16] ATLAS $\alpha(M_Z) = 127.95 \pm 0.017$.

A Second Variation Full Detailed

$$R = R_0 + \sum a_n \sin(n\theta + \pi\delta), A = \int \sqrt{R^2 + R'^2} d\theta \approx \int [R + R'^2/2R], \delta^2 A = \pi/2R_0 \sum n^2 a_n^2, \sum 1/(k + \delta)^2 = \pi^2 / \sin^2(\pi\delta), K_t = 1/|\sin(\pi\delta)|.$$

B Uehling Legacy Detailed

$$\Pi(q^2) = \alpha_0/\pi \int_0^1 dx x(1-x) \ln[1 + q^2/m^2 x(1-x)], V = e^2/r[1 + \alpha_0/\pi 1/15(r_e/r)^2], p_{\sigma_0} = (2\pi\sigma_0^2)^{-1/2} \exp[-\theta^2/2\sigma_0^2], U(0) = (2\sqrt{\pi}\sigma_0)^{-1}[\sigma_0^2 - 0.5\sigma_0^4], b_1 = \Phi^2/\sqrt{\pi}.$$